

CHP Fuel Supply — Background Notes for the Forest Association's Own Discussion

"The forest our ancestors planted — the power that sustains the village they built" 先人が植えた森が、今、村を支える力になる。

Status: everything below is a draft, and we are still finding a good way through this together — nothing here is decided, priced, or offered. This is not a term sheet from us to the co-op. It exists only to put some market numbers on the table so 地域林業パートナー — and Kinjo-san specifically — have something concrete to react to, question, or throw out entirely, ahead of the open item already flagged in

[mitsue_kanko_collaboration_diagrams.html](#) §"Discussion points." Every figure is illustrative, unconfirmed, and ours to revise once the co-op tells us what actually works for them.

0. Who leads what — the co-op is in the middle of this, not us

This project only works if the **Forest Association leads the forestry side**: how much gets harvested, at what pace, with what crew, and on what terms. That is their forest, their expertise, and their call. Our side (the mission entity / NGO, and the KK operating company it sets up and controls) is not trying to direct the co-op's work — our job is narrower:

- **Build and run the CHP + data center** so there is a reliable buyer for wood that currently has none.
- **Set up and operate the KK** to handle the financing, contracts, and revenue administration — the parts that are our expertise, not the co-op's.
- **Bring whatever market information we find** (prices, grades, benchmarks) to the table as input to the co-op's own decision, not as a price we're imposing.

Concretely, in the existing governance/labor diagrams ([mitsue_kanko_collaboration_diagrams.html](#) §§1, 3): the co-op plants, tends, harvests, chips, and dries wood — that whole chain is the co-op's operation. The KK's role is to buy what the co-op is willing to sell, on terms the co-op is willing to accept, and turn that into CHP fuel and DC power. If any of what follows reads like us setting the direction, that's a drafting problem to fix, not the intended relationship.

1. Some market context, for the co-op's own reference

Not an argument for a price — just numbers that might be useful background for however the co-op wants to think about this.

Current sugi stumpage is **¥4,127/m³ (March 2024), down from a ¥22,707/m³ peak in 1980, still falling** [林野庁 山元立木価格統計]. That price is for **A-grade sawlog** — straight, 16–36cm timber the sawmill wants. Most of what a **thinning** operation actually produces is not that grade. Per a Japanese thinning-site study (larch, Iwate Pref. Forestry Technology Center, 研究成果速報 No.308 — see [../energy-forest/workforce/mitsue_forest_workforce_energy_plan.md](#) §4a for full caveats — **this is not a Mitsue or sugi/hinoki study, treat as a rough reference only, and the co-op's own experience of what their thinnings actually yield should override it**):

Grade	Share of thinned volume (illustrative)	Existing market today
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Grade	Share of thinned volume (illustrative)	Existing market today
A材 (sawlog)	~22%	Yes — sawmill, at the stumpage price above
B材 (plywood-grade)	~54%	Sometimes — depends on a plywood/pulp buyer existing
C材 (fuel-grade, crooked)	~19%	Often none, as far as we know
D材 (tops/branches)	~5%	Often none, as far as we know

If this rough pattern holds for Mitsue's own stands — which only the co-op can actually confirm — it would suggest that a meaningful share of thinned wood (C+D, maybe more) doesn't have an existing buyer today, which is presumably part of why thinning has been hard to sustain. If the co-op's own experience says otherwise, that overrides everything in this table.

2. What the KK could put on the table, for the co-op to shape

Not an offer — a sketch of the *tools* the KK side could bring, for the co-op to push back on, reprice, or restructure however makes sense to them.

2a. A guaranteed, multi-year volume commitment

Something the open timber market can't offer: a standing buyer, over a term the co-op sets, at a volume that follows the co-op's own pace — not a target imposed from our side. If a multi-year commitment (illustratively 5–10 years) would actually help the co-op plan crew growth ([mitsue_forest_workforce_energy_plan.md](#) §1), we'd want to shape it around their schedule, not the other way around.

2b. A delivery rate — starting point only, co-op's to counter

The Mishima-derived benchmark used so far to calibrate the CHP model is a **≤¥7,000/m³ delivered chip cost ceiling** ([../meetings/forest_group_discussion_topics_jp.md](#)) — covering harvest + chip + dry + haul combined. Depending on who ends up handling which step, a starting number for discussion might land somewhere in the **¥3,000–5,000/m³** range for the co-op's own harvest-and-deliver work — but this is a number to react to, not a number we're proposing to hold to. If the co-op's real costs run higher, that's exactly the kind of thing this conversation needs to surface.

2c. A J-Credit share, as a separate strand

Current market ~¥3,000–6,000/t CO₂ ([forest_group_discussion_topics_jp.md](#) §credits), tied to the restoration side rather than the fuel price — one more thing to work out together, not a fixed add-on.

What this isn't: any claim on A-grade sawlog. That material should keep going wherever the co-op already sells it (or however they'd prefer to handle it) — the "Forestry products (timber, lumber)" line already in [mitsue_revenue_model.md](#) assumes exactly this, kept separate from CHP fuel.

3. Rough volume picture — not a target, just a scale check

Only useful for sanity-checking whether the CHP's fuel needs and the co-op's realistic capacity are even in the same ballpark. Not something we're asking the co-op to commit to.

	Total harvest/yr (if double-crew, per co-op's own plan)	Rough CHP-fuel-only floor (C+D, ~24%)	If B also has no local buyer (~78%)
Illustrative	42,000 m ³	~10,000 m ³	~32,800 m ³

Full table and caveats in `mitsue_forest_workforce_energy_plan.md` §4a. Whether B-grade has a local buyer is a question for the co-op to answer, not us — it changes this picture a lot.

3b. What this could mean for landowners — scale check, draft

Illustrative arithmetic only; nothing here is offered or committed (¥0 funded).

Item	Figure	Basis
50 kWe prototype fuel	~450–600 t dry/yr ≈ 1,400–1,900 m ³ solid/yr	<code>../energy-forest/chp-technical/mitsue_50kwe_prototype_fuel_demand.md</code>
Share of the C+D "fuel-only" floor (~10,000 m ³ /yr, §3)	~14–19%	§3 above
Electricity at 8,000 h	400 MWh/yr	50 kW × 8,000 h
FIT premium, plan-certified thinning wood (¥32 vs ¥24/kWh)	¥8/kWh ≈ ¥3.2M/yr	<code>../energy-forest/workforce/mitsue_forest_permit_subsidy_fit_reference.md</code>
Premium per m ³ if 100% passed upstream (ceiling)	≈ ¥1,700–2,300/m ³	¥3.2M ÷ 1,400–1,900 m ³

What this suggests:

- **Where landowners could gain:** a local buyer (and floor price) for C/D-grade wood that has none today; thinning pooled by the co-op across small plots; the plan-certified premium *if* the co-op files a 森林経営計画 and the premium is shared upstream.
- **Where it won't:** the 50 kWe prototype is too small to move local prices, and A/B sawlogs (most of landowner income) keep going to sawmills. Real price effect would need the ~2×0.6 MWe scale (Phase 4).
- **Caveats:** premium certification mechanics (self-cert vs audit) and eligibility for a ≤50 kWe unit are unverified — do not quote ¥8/kWh externally yet. A margin share depends on actual CHP revenue, which is unfunded.
- **Framing:** "an added buyer and premium for wood with no market today", complementary to the co-op and private harvesters — not "better than" any company.

4. Questions we don't have answers to — for the co-op to decide, not us

1. Does the co-op currently have a buyer for B-grade wood, or does it go unsold? (Changes the picture in §3 substantially — and only the co-op knows the real answer.)

2. What does the co-op's own harvest/chip/dry/haul cost actually look like per m³ today? Is ¥3,000–5,000/m³ anywhere close, or off?
3. Does the co-op's current crew support any near-term volume, or does hiring/mechanization need to come first — and on what timeline that works for them?
4. What contract term, volume ramp, and price structure would the co-op actually want — not what would suit us?
5. Species/replanting say — separate topic, tracked in `forest_group_discussion_topics_jp.md`, same principle: the co-op and NGO work this out together, not us deciding for them.

Sources

- Current stumpage price: 林野庁 山元立木価格統計 via 日本林業経済研究所 (shinrin-ringyou.com/data/kakaku_yama.php)
- Grade split: [岩手県林業技術センター 研究成果速報 No.308](#) (larch, illustrative only, not Mitsue-specific); [林野庁 2014年素材生産量統計](#) (national all-harvest average)
- Chip-cost benchmark: Ooba, Nakamura & Togawa (2020), Mishima CHP simulation, via `forest_group_discussion_topics_jp.md`
- Volume/crew scenarios: `mitsue_forest_workforce_energy_plan.md` §4, §4a
- Prefectural Forestry Corporation debt precedent (why the KK side is trying to avoid a debt-financed, timber-price-only model — not a reason to push a price onto the co-op): `mitsue_business_case.md` §2, Matsushita 2015, DOI 10.5772/61268